Fiscal Year:	FY 2017	**Task Last Updated:**	FY 08/11/2017
PI Name:	Minna, John D. M.D.		
Project Title:	NSCOR: Risks Estimates and Mechanisms of Lung Cancer Pathogenesis after Space Radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	john.minna@utsouthwestern.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	214 648-4900
Organization Name:	The University of Texas Southwestern Medical Center		
PI Address 1:	Hamon Center for Therapeutic Oncology Research		
PI Address 2:	6000 Harry Hines Boulevard, NB8.206		
PI Web Page:			
City:	Dallas	**State:**	TX
Zip Code:	75390-8593	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiation NSCOR/Virtual NSCOR NNJ10ZSA002N
Start Date:	01/01/2011	**End Date:**	12/31/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2016 per S. Monk/SR-LaRC (Ed., 10/20/15)		
Key Personnel Changes/Previous PI:	August 2017 report: For the final year of the grant we had added Dr. Rolf Brekken (UTSW) and Dr. Alan Balmain (UC San Francisco Medical Center) as key personnel (these were seen as CoInvestigators in the FY2016 report). We did this to help prepare for our competitive renewal. Each of them was to lead one of the two new projects and we required their input over the year to plan for this. No salary effort was given and no funds transferred for their use. Unfortunately, while the application received a fundable score it was not funded. When we learned this, we removed them as key personnel and focused on completing the previously approved projects.		
COI Name (Institution):	Aroumougame, Asaithamby Ph.D. (The University of Texas Southwestern Medical Center) Gazdar, Adi M.D. (The University of Texas Southwestern Medical Center) Girard, Luc Ph.D. (The University of Texas Southwestern Medical Center) Shay, Jerry Ph.D. (The University of Texas Southwestern Medical Center) Story, Michael Ph.D. (The University of Texas Southwestern Medical Center) Xie, Xian-Jin (The University of Texas Southwestern Medical Center) Xie, Yang Ph.D. (The University of Texas Southwestern Medical Center)		
Grant/Contract No.:	NNX11AC54G		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	This NASA Specialized Center of Research (NSCOR) focuses on the development of quantitative dose risk estimates following HZE (high energy) particle irradiation for the development of key genetic, epigenetic, gene expression, and cellular functional changes in the multi-step pathogenesis of lung cancer in both human bronchial epithelial cell (HBEC) and transgenic mouse models of lung cancer. The risk of developing these changes are being compared in these models to that of gamma-irradiation. These dose risk assessments are designed to measure events of both radiation-induced initiation and promotion events. They also measure effects on development of cancer stem-like cells (cancer initiating cells). 3D organotypic culture and animal models also allow measurement and risk estimation in both tissue surrogates and in vivo lung tissues. These individual risk estimates can then be combined into a model for overall risk of developing lung cancer from exposure to galactic cosmic radiation. To achieve these goals, University of Texas Southwestern Medical Center (UTSW) has assembled a team of scientists (Drs. John Minna, Jerry Shay, Michael Story, Yang Xie, Luc Girard, Adi Gazdar, D. Chen, and A. Asaithamby) who are leaders in the study and translation application of the molecular pathogenesis of lung cancer and in radiobiology. We have a panel of expert Internal and External Advisors (from other NSCORs and lung cancer research experts); as well as expert consultants and collaborators. This team has developed a novel immortalized human bronchial epithelial cell (HBEC) system that can be genetically manipulated, studied in 2D monolayer and 3D organotypic cultures. Using these HBECs we have completed many studies with HZE particle irradiation showing that further progression toward lung cancer can be detected and quantified and that specific mRNA expression profiles for HZE irradiation exist. In addition, we have found that transgenic mouse models of lung cancer have increased lung cancer risk after high LET (linear energy transfer) exposure allowing us to study the early onset and progression of lung cancer in the whole animal in real time. The NSCOR has 4 Projects: 1. HZE Particle Exposure and the Risk for Human Lung Carcinogenesis; 2. Mouse Models of Lung Cancer after HZE Particle Irradiation; 3. Organotypic (3D) Human Models of Genomic Instability for HZE Particle Induced Lung Carcinogenesis; 4. Integrating Biomarkers of Lung Cancer Pathogenesis for Risk Assessment after HZE Particle Irradiation. These projects are supported by 2 Cores: Administrative Core and the Database, Biostatistics and Bioinformatics Core. Our NSCOR has substantial UTSW institutional commitments, a recently renewed National Cancer Institute (NCI) Special Program of Research Excellence (SPORE) in Lung Cancer, a prior Center for the NCI Early Detection Research Network (EDRN) program, and key investigators that are part of other funded NSCORs (Shay and Story). These provide great synergism and additional resources available for the successful completion of this renewal proposal.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Lung cancer is the leading cause of cancer death in both women and men in the USA and in the Western world. While lung cancer is caused mostly by smoking exposure, a significant number of lung cancers arise in lifetime never smokers, and thus is a sizable health problem in the USA and world-wide. Indeed lung cancer in never smokers is the 7th leading cause of world-wide cancer related deaths. Likewise, 50% of all new lung cancer cases occur in former smokers (quitting over 5 years before). In all of these scenarios a major underlying question has been the role of other environmental carcinogens especially environmental radiation. One source of this is radon gas in the environment including radon gas exposure in homes with alpha particles (which represent high-LET radiation). Thus, the interaction of radiation in the environment on Earth including from alpha particles in the genesis of lung cancer and quantitation of the effect of such radiation on lung epithelial cells is of major health importance. Radiation is a known carcinogenic influence but the molecular events in lung epithelial cells in response to radiation still remains to be determined. A key element of this is the impact of radiation on lung epithelial cells with preneoplastic lesions.
	Progress Report (2015-2016): Project 2: Mouse Models of Lung and Other Cancers after Space Radiation: (J.W. Shay, Lead). Space radiation poses a major risk to human health during extended space mission. Galactic Cosmic Radiation (GCR) consists of 2% electrons, 85.3% protons (hydrogen nuclei), 11.8% alpha particles (nuclei), and less than 1% of high charge (Z) and energy E (HZE) particles. Protons account for the vast majority of radiation in the space in the form of a Solar Particle Events (SPEs). SPEs are comprised of varying energies and doses over a period of time (protracted dose of radiation), and occurrences are difficult to predict. While potentially most of this radiation can be reduced by shielding, the biological consequences of constant exposure to even small amount of proton irradiation remains unknown It is predicted that SPEs have high carcinogenic effects compared to equivalent low energy terrestrial radiation (e.g., X-rays). However, data are still required to determine more exactly the increased risk of invasive cancer with low dose rates and varying energies of protons. Our lab is interested in understanding the biological effects such an increase has in progression of invasive carcinoma in the lung due to the various energies and dose-rates. Based on analysis of human populations exposed to radiation, the lung is susceptible to tumor development following exposure. For these studies, we used the K-rasLA1 mouse model which mimics the development of human adenocarcinoma non-small cell lung cancer progression by spontaneous activation of mutant K-ras lesions. Using the K-rasLA1 mouse model, we studied overall survival and the progression of lung cancer after total body exposure to a simulated SPE with varying energies (50 – 150 MeV/n) using a total dose 0.5 Gy, 1.0 Gy, and 2.0 Gy (at an average dose rate of 1.67 cGy/min). We also exposed mice to acute proton doses at (50 and 150 MeV/n) and to 2 Gy of X-rays (250 kVp). The SPEs simulation, and monoenergetic proton radiation lead to increases in induced invasive carcinomas as compared to X-ray radiation. Even comparing 0.5 Gy of protons (as a SPEs) we observed increased carcinoma incidence compared to 2 Gy of X-rays. In addition, K-rasLA1 mice exposed to 2.0 Gy of sSPE radiation and 2.0 Gy of monoenergetic acute proton (50 MeV/n or 150 MeV/n) exhibited a significant decrease in median survival compared to unirradiated control mice. To evaluate the underlying mechanistic details involved in radiation-mediated tumorigenesis in our lung cancer model, the phosphorylation status (activation) of various targets important to the processes of tumorigenesis were investigated.

Task Progress:	We found K-rasLA1 mice exposed to energetic protons exhibited altered growth factor signaling compared to unirradiated controls. We hypothesize that alteration in growth factors are due to the chronic oxidative stress caused by the SPEs leading to increase in invasive cancer. Further studies are in progress to understand the SPEs biological effect. Significant biological and mechanistically data obtained from these studies may help in risk assessment of space travel and provide insights into molecular mechanisms which could be applicable in mitigating or preventing cancer initiation and progress during long-duration space travel. In addition, we found that the countermeasure CDDO, an anti-inflammatory and anti-oxidant, provided orally in mouse lab chow, greatly reduced the initiation and progression of lung cancer in the K-rasLA1 mouse model without radiation. CDDO is Bardoxolone methyl, also known as “RTA 402”, “CDDO-methyl ester”, and CDDO-Me) is an experimental and orally-bioavailable semi-synthetic triterpenoid, based on the scaffold of the natural product oleanolic acid. Preclinical studies indicate that CDDO-Me acts as an activator of the Nrf2 pathway and an inhibitor of the NF-kappaB pathway. Summary of CDDO Countermeasure Experiments • 2.0 Gy SPE simulation and 1.0 Gy fractionated 56Fe- decreases the lifespan of K-rasLA1 animals and results in an increase incidence of invasive carcinoma. K-rasLA1 mice on a CDDO diet only during fractionated 56Fe- irradiation exhibit a 2-fold decreased incidence in invasive carcinoma in comparison to mice on control diet with no side effects. • K-rasLA-1 mice on CDDO diet during solar particle irradiation simulations exhibit a decreased incidence in invasive carcinoma in comparison to mice on control diet. • K-rasLA-1 mice fed CDDO without irradiation show reduced tumor progression. • There were no hematological or other tissue toxicities even when mice were kept on CDDO for long time periods (>100 days). Core A: Administration (J. Minna, Core Lead; J. Shay, Core Co-Lead; Administrative Lead, M. Warwick). The purpose of the administrative core during this no cost extension year was to continue to provide administrative, fiscal, and management support for the UT Southwestern Lung Cancer NSCOR. This core provided the coordination of travel for experiments at BNL. Ms. Warwick served as Chief Administrative Officer of the NSCOR. Core B: NSCOR Computational Biology and Pathology Support Core (A Gazdar, Co-Lead). During the no-cost extension year the pathology core provided histopathology review support for NSCOR project 2.
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